USER ACCEPTANCE TESTING (UAT) FORM

TBhon: IoT-Enabled Mobile-Based Multimodal TB Pre-Screening System

Instructions: Complete this form after performing the UAT tasks below using the TBhon mobile application. Rate each criterion based on your professional experience. One form per evaluator.

Section A: Evaluator Information

Name:	____
Role / Position:	____
Organization:	____
Date:	____

Section B: System Information

System Name:	TBhon
Version:	1.2 (Prototype / Field Evaluation Build)
Purpose:	To support tuberculosis pre-screening and triage using an 11-item symptom checklist, CNN-based cough audio analysis (Mel-spectrogram), sputum smear image classification, weighted multimodal risk fusion (Low / Moderate / High), optional ESP32 IoT capture, and cloud-backed session management via React Native mobile app, Express/Prisma backend, and FastAPI ML inference service.
Scope Note:	TBhon is triage support only — not a medical diagnosis. Standard clinical evaluation and laboratory testing are required for diagnosis and treatment decisions.

Section B.1: UAT Tasks (complete before rating)

1. 1. Register or log in as booth staff and start a new walk-in screening session.
2. 2. Complete client intake and the 11-item TB symptom checklist.
3. 3. Record or upload cough audio (mobile mic or ESP32 IoT module, if available).
4. 4. Capture or upload a sputum smear image (camera or ESP32 microscope module, if available).
5. 5. Review ML processing status, staff-review screen, and fused triage risk result (Low / Moderate / High).
6. 6. Verify screening history, patient QR result claim (if applicable), and disclaimer visibility.

Section C: Evaluation Criteria (ISO/IEC 25010 Software Quality Model)

No.	Criteria	ISO/IEC 25010	5	4	3	2	1	Remarks
1	The system supports the complete TB pre-screening workflow (client intake, checklist, cough capture, sputum capture, staff review, and result disclosure).	Functional Suitability						
2	The 11-item symptom checklist captures relevant TB screening indicators for triage.	Functional Suitability						
3	Cough audio analysis (CNN / Mel-spectrogram) produces useful and interpretable triage signals.	Functional Suitability						
4	Sputum smear image analysis produces useful and interpretable triage signals.	Functional Suitability						
5	Multimodal risk fusion (Low, Moderate, High) is appropriate for preliminary TB triage support.	Functional Suitability						
6	Screening reports, recommendations, and referral guidance are clear and actionable.	Functional Suitability						
7	Patient result access (QR claim) and screening history functions work as intended.	Functional Suitability						
8	Cough audio classification completes within an acceptable time (target: ≤ 5 seconds).	Performance Efficiency						
9	Sputum image classification completes within an acceptable time (target: ≤ 5 seconds).	Performance Efficiency						
10	The mobile application responds smoothly during routine screening tasks.	Performance Efficiency						
11	ML inference and cloud backend connectivity perform adequately during testing.	Performance Efficiency						
12	The application is easy to use for booth staff with minimal technical training.	Usability						
13	The interface design is clear, consistent, and appropriate for a health screening context.	Usability						
14	Navigation through the staff-guided screening flow is intuitive and logical.	Usability						
15	Triage disclaimers (not a diagnosis) are visible, understandable, and appropriately placed.	Usability						
16	Processing status and error guidance during ML analysis are clear and helpful.	Usability						
17	The system operates reliably without unexpected crashes or data loss during screening.	Reliability						
18	Screening session data is stored correctly and can be retrieved from history.	Reliability						
19	ESP32 IoT cough/sputum capture integrates reliably when used (if tested).	Reliability						
20	The system handles unstable network conditions in a predictable manner.	Reliability						
21	User authentication (login, registration, password policy) protects system access appropriately.	Security						
22	Patient screening data is handled with appropriate privacy and confidentiality.	Security						
23	Cloud-backed data storage (Express/Prisma backend) is trustworthy for screening records.	Security						
24	Overall, I am satisfied with TBhon as a TB pre-screening and triage support tool.	Overall Satisfaction						

Rating Scale: 5 – Strongly Agree | 4 – Agree | 3 – Neutral | 2 – Disagree | 1 – Strongly Disagree

Section D: Overall Rating

How would you rate the overall performance of TBhon?

☐ Excellent	☐ Good	☐ Fair	☐ Poor	☐ Very Poor

Section E: Comments / Suggestions

Please provide any additional feedback on usability, accuracy, reliability, security, or features that could improve TBhon for community TB screening booths.

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Section F: Evaluator Confirmation

Signature:	Date:
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Target Respondents: *Community health workers, nurses, public health professionals, and IT experts (purposive sampling per study methodology).*